

MATH-3108-103-71037-2022FS-Linear Algebra I Exam 3

McLain Oertle

TOTAL POINTS

94 / 100

QUESTION 1

1 Question 1 0 / 3

✓ - **3 pts** Incorrect. The correct answer is 3.

QUESTION 2

2 Question 2 3 / 3

✓ - **0 pts** Correct

QUESTION 3

3 Question 3 3 / 3

✓ - **0 pts** Correct

QUESTION 4

4 Question 4 3 / 3

✓ - **0 pts** Correct

QUESTION 5

5 Question 5 3 / 3

✓ - **0 pts** Correct

QUESTION 6

Question 6 6 pts

6.1 Question 6 - first part 3 / 3

✓ - **0 pts** Correct

6.2 Question 6 - second part 3 / 3

✓ - **0 pts** Correct

QUESTION 7

7 Question 7 9 / 9

✓ - **0 pts** Correct

QUESTION 8

8 Question 8 10 / 10

✓ - **0 pts** Correct

QUESTION 9

9 Question 9 14 / 14

✓ - **0 pts** Correct

QUESTION 10

Question 10 8 pts

10.1 Question 10a 5 / 6

✓ - **1 pts** Answer not simplified or simplified incorrectly. The simplified form of the answer is $10\sqrt{2}$.

10.2 Question 10b 2 / 2

✓ - **0 pts** Correct

QUESTION 11

11 Question 11 7 / 9

✓ - **2 pts** Two incorrect entries. The correct answer is $\left[\begin{array}{ccc} c & c & c \end{array} \right] \{ 3 \} \& \{ 0 \} \& \{ 2 \} \setminus \{ 0 \} \& \{ 2 \} \& \{ -3 \} \setminus \{ 2 \} \& \{ -3 \} \& \{ -5 \} \end{array} \right]$.

QUESTION 12

12 Question 12 10 / 10

✓ - 0 pts Correct

QUESTION 13

13 Question 13 3 / 3

✓ - 0 pts Correct

QUESTION 14

14 Question 14 16 / 16

✓ - 0 pts Correct

Name: McLain Oertle

Math 3108 – Linear Algebra
Exam 3
Prof. Runnion
December 5, 2022

Directions:

- 1) All cell phones and other electronic noisemaking devices must be turned completely off (silence or on vibrate is *not* acceptable) and put away for the duration of the exam.
- 2) No calculators, books, or other materials are permitted.
- 3) You must write **darkly** and **legibly** – this exam will be scanned for electronic grading.
- 4) Show **ALL** your work! Correct answers which are not properly justified will not receive full credit.
- 5) Failure to follow directions specific to a problem will result in the loss of points.
- 6) Circle, box, or underline each final answer if necessary for clarity. Label all graphs.
- 7) If you work a problem two different ways, clearly indicate which one you want us to grade, preferably by crossing out the one you do not want us to look at. If you use two methods, one of which is wrong, and neither method is crossed out, you will not receive full credit.
- 8) Answers must be exact (like $\sqrt{2}$), not approximate (like 1.414), unless a problem specifically indicates otherwise.
- 9) Simplify where appropriate. Quantities such as $\sqrt{9}$ and $\cos \pi$ should be calculated.
- 10) Extra space has been provided on all problems. There should be more than sufficient space to work all problems in the space indicated. Thus, NO WORK APPEARING ON THE BACK OF PAGES WILL BE GRADED!
- 11) All work must appear within this exam packet to be graded. **No scratch paper is allowed under any circumstances.**
- 12) This packet has six sheets of paper, including this cover page. Do NOT remove the staple or remove any sheet from this packet.
- 13) Once this exam begins, you will have 50 minutes to complete your solutions.

**DO NOT OPEN THIS EXAM
UNTIL TOLD TO DO SO**

Name: McLain Oerthe

Multiple Choice and True/False. Completely fill in the box corresponding to the correct answer(s). On the multiple choice problems, assume all matrices and vectors contain only real entries. 3 points each.

1. Let A be a 5×5 **symmetric** matrix and let λ be an eigenvalue of A with multiplicity 3. The **minimum** dimension of the eigenspace of A corresponding to λ is _____.

☒ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

2. Every orthogonal set of vectors is linearly independent.

☐ a) True ☒ b) False

3. Let W be a real vector space. If $\text{proj}_W \mathbf{y} = \mathbf{y}$, then $\mathbf{y}$ is in W .

☒ a) True ☐ b) False

4. The set of all vectors in $\mathbb{R}^n$ orthogonal to one fixed nonzero vector is a subspace of $\mathbb{R}^n$.

☒ a) True ☐ b) False

5. An $m \times n$ matrix U has _____ columns if and only if $U^T U = I$.

☐ a) Orthogonal ☒ b) Orthonormal

6. Consider the matrix $A = \begin{bmatrix} 1 & 2 & 4 & 3 \\ 0 & 5 & 7 & 9 \\ 0 & 0 & 7 & 6 \\ 0 & 0 & 0 & 8 \end{bmatrix}$

(3 points) Is A diagonalizable? ☒ Yes ☐ No

(3 points) How many distinct eigenvalues does A have?

☐ 0 ☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5 ☐ 6

Name: Melvin Perle

Open Answer. Show ALL your work!

7. (9 points) Let $W = \text{Span}\{\mathbf{u}_1, \mathbf{u}_2\}$, where $\mathbf{u}_1 = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}$ and $\mathbf{u}_2 = \begin{bmatrix} 5 \\ 1 \\ 4 \end{bmatrix}$, and let $\mathbf{y} = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$.

Write $\mathbf{y}$ as the sum of a vector $\hat{\mathbf{y}}$ in W and a vector $\mathbf{z}$ orthogonal to W .

$$\hat{\mathbf{y}} = \frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \mathbf{u}_1 + \frac{\mathbf{y} \cdot \mathbf{u}_2}{\mathbf{u}_2 \cdot \mathbf{u}_2} \mathbf{u}_2 = \frac{1+9-10}{1+9-10} \mathbf{u}_1 + \frac{5+3+20}{25+1+16} \mathbf{u}_2 = \frac{28}{42} = \frac{14}{21} \mathbf{u}_2 = \frac{2}{3} \mathbf{u}_2 = \frac{2}{3} \begin{bmatrix} 5 \\ 1 \\ 4 \end{bmatrix}$$

$$\hat{\mathbf{y}} = \begin{bmatrix} \frac{10}{3} \\ \frac{2}{3} \\ \frac{8}{3} \end{bmatrix}$$

$$\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}} = \begin{bmatrix} \frac{3}{3} \\ \frac{7}{3} \\ \frac{15}{3} \end{bmatrix} = \begin{bmatrix} 1 \\ \frac{7}{3} \\ 5 \end{bmatrix}$$

$$\mathbf{y} = \begin{bmatrix} \frac{10}{3} \\ \frac{2}{3} \\ \frac{8}{3} \end{bmatrix} + \begin{bmatrix} 1 \\ \frac{7}{3} \\ 5 \end{bmatrix}$$

8. (10 points) Find the eigenvalues and associated eigenvectors of $A = \begin{bmatrix} 0 & 1 \\ -8 & 4 \end{bmatrix}$

$$A - \lambda I = \begin{bmatrix} -\lambda & 1 \\ -8 & 4-\lambda \end{bmatrix} = -\lambda(4-\lambda) + 8 = \lambda^2 - 4\lambda + 8 = \frac{4 \pm \sqrt{16-32}}{2} = \frac{4 \pm 4i}{2} = 2 \pm 2i$$

$$\lambda = 2+2i, 2-2i$$

$$\begin{bmatrix} -2-2i & 1 \\ -8 & 2-2i \end{bmatrix} = 0 \rightarrow \begin{bmatrix} -2-2i & 1 & 0 \\ -8 & 2-2i & 0 \end{bmatrix} \sim \begin{bmatrix} -8 & 2-2i & 0 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & \frac{-2+2i}{8} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & \frac{1}{4} + \frac{1}{4}i & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{eigenvector} = \begin{bmatrix} \frac{1}{4} - \frac{1}{4}i \\ 1 \end{bmatrix} \text{ and complex conjugate} = \begin{bmatrix} \frac{1}{4} + \frac{1}{4}i \\ 1 \end{bmatrix}$$

Name: McLain Dertle

9. (14 points) Orthogonally diagonalize $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$, if possible. If it is not possible, clearly explain why.

$$\begin{bmatrix} 3-\lambda & 1 \\ 1 & 3-\lambda \end{bmatrix} \rightarrow (3-\lambda)(3-\lambda) - 1 = 9 - 6\lambda + \lambda^2 - 1 = \lambda^2 - 6\lambda + 8 = (\lambda-2)(\lambda-4)$$

$\lambda = 2, 4$

$\lambda = 2$

$$\begin{bmatrix} 3-2 & 1 \\ 1 & 3-2 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

A_1, B_1 x_2 is free

$$x_1 = -x_2 \quad \vec{v} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$\lambda = 4$

$$\begin{bmatrix} 3-4 & 1 \\ 1 & 3-4 \end{bmatrix} \sim \begin{bmatrix} -1 & 1 & 0 \\ 1 & -1 & 0 \end{bmatrix} \sim \begin{bmatrix} -1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\lambda_1 = \lambda_2 \rightarrow \vec{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$$D = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \quad Q = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$$

10. a) (6 points) Find the distance between $\mathbf{u} = \begin{bmatrix} 12 \\ 3 \\ -5 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 2 \\ -3 \\ 3 \end{bmatrix}$.

$$\|\vec{u} - \vec{v}\| = \left\| \begin{bmatrix} 12 \\ 3 \\ -5 \end{bmatrix} - \begin{bmatrix} 2 \\ -3 \\ 3 \end{bmatrix} \right\| = \left\| \begin{bmatrix} 10 \\ 6 \\ -8 \end{bmatrix} \right\| = \sqrt{100 + 36 + 64} = \sqrt{200}$$

$$\text{distance} = \sqrt{200}$$

- b) (2 points) Are $\mathbf{u}$ and $\mathbf{v}$ orthogonal? ☒ Yes ☐ No (fill in the box)

orthogonal if $\mathbf{u} \cdot \mathbf{v} = 0$

$$24 - 9 - 15 = 0$$

Name: McLain Oerlke

11. (9 points) Find the matrix of the quadratic form. Assume $\mathbf{x}$ is in $\mathbb{R}^3$.

$$3x_1^2 + 2x_2^2 - 5x_3^2 + 4x_1x_3 - 6x_2x_3$$

0 because no x_1x_2 term

$$A = \begin{bmatrix} 3 & 0 & 2 \\ 0 & 2 & -3 \\ 2 & -3 & -5 \end{bmatrix}$$

12. (10 points) Given the matrix $A = \begin{bmatrix} 5 & -2 & 6 & -1 \\ 0 & 3 & h & 0 \\ 0 & 0 & 5 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$, find the value for h such that the

eigenspace corresponding to $\lambda = 5$ is two-dimensional.

R_1, R_3

$$\begin{bmatrix} 0 & -2 & 6 & -1 \\ 6 & -2 & h & 0 \\ 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & -4 \end{bmatrix} \xrightarrow{R_1, R_3} \begin{bmatrix} 0 & -2 & 6 & -1 \\ 0 & -2 & h & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

if $h \neq 6$ then 1 dimensional

$$\begin{bmatrix} 0 & 2 & 6 & -1 \\ 0 & -2 & 6 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{-R_1, R_2} \begin{bmatrix} 0 & 2 & 6 & -1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

λ free $\rightarrow 2D$
 λ_3 free

$$\boxed{h=6}$$

13. (3 points) Find $\|\mathbf{v}\|$ given $\mathbf{v} = \begin{bmatrix} -1 \\ 2 \\ 0 \\ -2 \end{bmatrix}$.

$$\|\mathbf{v}\| = \sqrt{(-1)^2 + 2^2 + 0^2 + (-2)^2} = \sqrt{1+4+0+4} = \sqrt{9} = 3$$

$$\|\mathbf{v}\| = \boxed{3}$$

Name: Adrian Perle

14. (16 points) The vectors $\mathbf{x}_1 = \begin{bmatrix} 0 \\ 4 \\ 2 \end{bmatrix}$ and $\mathbf{x}_2 = \begin{bmatrix} 5 \\ 6 \\ -7 \end{bmatrix}$ form a basis for a subspace W of $\mathbb{R}^3$.

Use the Gram-Schmidt process to produce an orthogonal basis $\{\mathbf{v}_1, \mathbf{v}_2\}$ for W where $\mathbf{x}_1 = \mathbf{v}_1$. Then, construct an orthonormal basis $\{\mathbf{u}_1, \mathbf{u}_2\}$.

$$\mathbf{v}_1 = \mathbf{x}_1$$

$$\mathbf{v}_2 = \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 = \begin{bmatrix} 5 \\ 6 \\ -7 \end{bmatrix} - \frac{0+24-14}{0+16+4} \begin{bmatrix} 0 \\ 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \\ -7 \end{bmatrix} - \frac{10}{20} \begin{bmatrix} 0 \\ 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \\ -7 \end{bmatrix} - \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}$$

$$\mathbf{v}_2 = \begin{bmatrix} 5 \\ 4 \\ -8 \end{bmatrix}$$

$$\text{orthogonal basis } \{\vec{\mathbf{v}}_1, \vec{\mathbf{v}}_2\} = \left\{ \begin{bmatrix} 0 \\ 4 \\ 2 \end{bmatrix}, \begin{bmatrix} 5 \\ 4 \\ -8 \end{bmatrix} \right\}$$

orthonormal basis:

$$\frac{\vec{\mathbf{v}}_1}{\|\vec{\mathbf{v}}_1\|} = \frac{\vec{\mathbf{v}}_1}{\sqrt{0+16+4}} = \frac{\vec{\mathbf{v}}_1}{\sqrt{20}} = \frac{\vec{\mathbf{v}}_1}{2\sqrt{5}}$$

$$\mathbf{u}_1 = \begin{bmatrix} 0 \\ \frac{4}{2\sqrt{5}} \\ \frac{2}{2\sqrt{5}} \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix}$$

$$\frac{\vec{\mathbf{v}}_2}{\|\vec{\mathbf{v}}_2\|} = \frac{\vec{\mathbf{v}}_2}{\sqrt{25+16+64}} = \frac{\vec{\mathbf{v}}_2}{\sqrt{105}}$$

$$\mathbf{u}_2 = \begin{bmatrix} \frac{5}{\sqrt{105}} \\ \frac{4}{\sqrt{105}} \\ \frac{-8}{\sqrt{105}} \end{bmatrix}$$

$$\text{orthonormal basis } \{\vec{\mathbf{u}}_1, \vec{\mathbf{u}}_2\} = \left\{ \begin{bmatrix} 0 \\ \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix}, \begin{bmatrix} \frac{5}{\sqrt{105}} \\ \frac{4}{\sqrt{105}} \\ \frac{-8}{\sqrt{105}} \end{bmatrix} \right\}$$